



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Title of Thesis

Master Thesis

S. Tudent

January 19, 2038

Advisors: Dr. R. Käppeli

Seminar for Applied Mathematics,
Department of Mathematics, ETH Zürich

Abstract

This example thesis briefly shows the main features of our thesis style, and how to use it for your purposes.

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Chapter 1

Introduction

This is a slightly adapted¹ version v1.4 of the template from <https://www.cadmo.ethz.ch/education/thesis/template.html>.

We assume that you found this template on our institute's website, so we do not repeat everything stated there. Consult the website again for pointers to further reading about L^AT_EX. This chapter only gives a brief overview of the files you are looking at.

1.1 Features

The rest of this document shows off a few features of the template files. Look at the source code to see which macros we used!

The template is divided into T_EX files as follows:

1. `thesis.tex` is the main file.
2. `extrapackages.tex` holds extra package includes.
3. `layoutsetup.tex` defines the style used in this document.
4. `theoremsetup.tex` declares the theorem-like environments.
5. `macrosetup.tex` defines extra macros that you may find useful.
6. `introduction.tex` contains this text.
7. `sections.tex` is a quick demo of each sectioning level available.
8. `declaration-originality.pdf` is included at the end of the thesis. However, simply filling out the form and including it will not work. To include it, print the filled out form using a PDF printer and name it `declaration-originality.pdf`².

¹E.g., we are using *cleveref* instead of *varioref*.

²You can find the original in `declaration-originality.original.pdf`.

9. `refs.bib` is an example bibliography file containing all the references we've discussed so far. You can use Bib $\TeX$ to quote references. For example, we have discussed [6, 3, 4, 5].
10. Here is an example of citing a lecture [2].
11. Here is an example of citing a code repository [1].

1.1.1 Extra package includes

The file `extrapackages.tex` lists some packages that usually come in handy. Simply have a look at the source code. We have added the following comments based on our experiences:

REC This package is recommended.

OPT This package is optional. It usually solves a specific problem in a clever way.

ADV This package is for the advanced user, but solves a problem frequent enough that we mention it. Consult the package's documentation.

As a small example, here is a reference to the Section *Features* typeset with the recommended *cleveref* package:

See Section 1.1.

See Table 2.1.

See Eq. (1.1).

Here is Euler's identity so that we can reference it above

$$e^{i\pi} + 1 = 0. \tag{1.1}$$

The same works for figures, listings, etc.

1.1.2 Layout setup

This defines the overall look of the document – for example, it changes the chapter and section heading appearance. We consider this a 'do not touch' area. Take a look at the excellent *Memoir* documentation before changing it.

In fact, take a look at the excellent *Memoir* documentation, full stop.

1.1.3 Theorem setup

This file defines a bunch of theorem-like environments.

Theorem 1.1 *An example theorem.*

Proof Proof text goes here. □

Note that the q.e.d. symbol moves to the correct place automatically if you end the proof with an `enumerate` or `displaymath`. You do not need to use `\qedhere` as with *amsthm*.

Theorem 1.2 (Some Famous Guy) *Another example theorem.*

Proof This proof

1. ends in an `enumerate`. □

Proposition 1.3 *Note that all theorem-like environments are by default numbered on the same counter.*

Proof This proof ends in a display like so:

$$f(x) = x^2. \quad \square$$

1.1.4 Macro setup

For now the macro setup only shows how to define some basic macros, and how to use a neat feature of the *mathtools* package:

$$|a|, \quad \left|\frac{a}{b}\right|, \quad \left|\frac{a}{b}\right|.$$

Chapter 2

Writing scientific texts in English

This chapter was originally a separate document written by Reto Spöhel. It is reprinted here so that the template can serve as a quick guide to thesis writing, and to provide some more example material to give you a feeling for good typesetting.

2.1 Basic writing rules

The following rules need little further explanation; they are best understood by looking at the example in the booklet by Knuth et al., §2–§3.

Rule 2.1 Write texts, not chains of formulas.

More specifically, write full sentences that are logically interconnected by phrases like ‘Therefore’, ‘However’, ‘On the other hand’, etc. where appropriate.

Rule 2.2 Displayed formulas should be embedded in your text and punctuated with it.

In other words, your writing should not be divided into ‘text parts’ and ‘formula parts’; instead the formulas should be tied together by your prose such that there is a natural flow to your writing.

2.2 Being nice to the reader

Try to write your text in such a way that a reader enjoys reading it. That’s of course a lofty goal, but nevertheless one you should aspire to!

Rule 2.3 Be nice to the reader.

Give some intuition or easy example for definitions and theorems which might be hard to digest. Remind the reader of notations you introduced

many pages ago – chances are he has forgotten them. Illustrate your writing with diagrams and pictures where this helps the reader. Etc.

Rule 2.4 Organize your writing.

Think carefully about how you subdivide your thesis into chapters, sections, and possibly subsections. Give overviews at the beginning of your thesis and of each chapter, so the reader knows what to expect. In proofs, outline the main ideas before going into technical details. Give the reader the opportunity to ‘catch up with you’ by summing up your findings periodically.

Useful phrases: ‘So far we have shown that ...’, ‘It remains to show that ...’, ‘Recall that we want to prove inequality (7), as this will allow us to deduce that ...’, ‘Thus we can conclude that Next, we would like to find out whether ...’, etc.

Rule 2.5 Don’t say the same thing twice without telling the reader that you are saying it twice.

Repetition of key ideas is important and helpful. However, if you present the same idea, definition or observation twice (in the same or different words) without telling the reader, he will be looking for something new where there is nothing new.

Useful phrases: ‘Recall that [we have seen in Chapter 5 that] ...’, ‘As argued before / in the proof of Lemma 3, ...’, ‘As mentioned in the introduction, ...’, ‘In other words, ...’, etc.

Rule 2.6 Don’t make statements that you will justify later without telling the reader that you will justify them later.

This rule also applies when the justification is coming right in the next sentence! The reasoning should be clear: if you violate it, the reader will lose valuable time trying to figure out on his own what you were going to explain to him anyway.

Useful phrases: ‘Next we argue that ...’, ‘As we shall see, ...’, ‘We will see in the next section that ...’, etc.

2.3 A few important grammar rules

Rule 2.7 There is (almost) *never* a comma before ‘that’.

It’s really that simple. Examples:

We assume that ...
Wir nehmen an, dass ...

It follows that ...

Daraus folgt, dass ...

'thrice' is a word that is seldom used.

'thrice' ist ein Wort, das selten verwendet wird.

Exceptions to this rule are rare and usually pretty obvious. For example, you may end up with a comma before 'that' because 'i.e.' is spelled out as 'that is':

For $p(n) = \log n/n$ we have ... However, if we choose p a little bit higher, that is $p(n) = (1 + \varepsilon) \log n/n$ for some $\varepsilon > 0$, we obtain that...

Or you may get a comma before 'that' because there is some additional information inserted in the middle of your sentence:

Thus we found a number, namely n_0 , that satisfies equation (13).

If the additional information is left out, the sentence has no comma:

Thus we found a number that satisfies equation (13).

(For 'that' as a relative pronoun, see also Rules 2.9 and 2.10 below.)

Rule 2.8 There is usually no comma before 'if'.

Example:

A graph is not 3-colorable if it contains a 4-clique.

Ein Graph ist nicht 3-färbbar, wenn er eine 4-Clique enthält.

However, if the 'if' clause comes first, it is usually separated from the main clause by a comma:

If a graph contains a 4-clique, it is not 3-colorable .

Wenn ein Graph eine 4-Clique enthält, ist er nicht 3-färbbar.

There are more exceptions to these rules than to Rule 2.7, which is why we are not discussing them here. Just keep in mind: don't put a comma before 'if' without good reason.

Rule 2.9 Non-defining relative clauses have commas.

Rule 2.10 Defining relative clauses have no commas.

In English, it is very important to distinguish between two types of relative clauses: defining and non-defining ones. This is a distinction you absolutely need to understand to write scientific texts, because mistakes in this area actually distort the meaning of your text!

It's probably easier to explain first what a *non-defining* relative clause is. A non-defining relative clauses simply gives additional information *that could also be left out* (or given in a separate sentence). For example, the sentence

The WEIRDSORT algorithm, which was found by the famous mathematician John Doe, is theoretically best possible but difficult to implement in practice.

would be fully understandable if the relative clause were left out completely. It could also be rephrased as two separate sentences:

The WEIRDSORT algorithm is theoretically best possible but difficult to implement in practice. [By the way,] WEIRDSORT was found by the famous mathematician John Doe.

This is what a non-defining relative clause is. *Non-defining relative clauses are always written with commas.* As a corollary we obtain that you cannot use ‘that’ in non-defining relative clauses (see Rule 2.7!). It would be wrong to write

~~The WEIRDSORT algorithm, that was found by the famous mathematician John Doe, is theoretically best possible but difficult to implement in practice.~~

A special case that warrants its own example is when ‘which’ is referring to the entire preceding sentence:

Thus inequality (7) is true, which implies that the Riemann hypothesis holds.

As before, this is a non-defining relative sentence (it could be left out) and therefore needs a comma.

So let’s discuss *defining* relative clauses next. A defining relative clause tells the reader *which specific item the main clause is talking about*. Leaving it out either changes the meaning of the sentence or renders it incomprehensible altogether. Consider the following example:

The WEIRDSORT algorithm is difficult to implement in practice.
In contrast, the algorithm that we suggest is very simple.

Here the relative clause ‘that we suggest’ cannot be left out – the remaining sentence would make no sense since the reader would not know which algorithm it is talking about. This is what a defining relative clause is. *Defining relative clauses are never written with commas.* Usually, you can use both ‘that’ and ‘which’ in defining relative clauses, although in many cases ‘that’ sounds better.

As a final example, consider the following sentence:

For the elements in $\mathcal{B}$ which satisfy property (A), we know that equation (37) holds.

2.4. Things you (usually) don't say in English

Table 2.1: Things you (usually) don't say

It holds (that) ... (‘Equation (5) holds.’ is fine, though.)	We have ...	<i>Es gilt ...</i>
x fulfills property $\mathcal{P}$.	x satisfies property $\mathcal{P}$.	<i>x erfüllt Eigenschaft $\mathcal{P}$.</i>
in average	on average	<i>im Durchschnitt</i>
estimation	estimate	<i>Abschätzung</i>
composed number	composite number	<i>zusammengesetzte Zahl</i>
with the help of	using	<i>mit Hilfe von</i>
surely	clearly	<i>sicher, bestimmt</i>
monotonously increasing	monotonically incr.	<i>monoton steigend</i>
(Actually, in most cases ‘increasing’ is just fine.)		

This sentence does not make a statement about all elements in $\mathcal{B}$, only about those satisfying property (A). The relative clause is *defining*. (Thus we could also use ‘that’ in place of ‘which’.)

In contrast, if we add a comma the sentence reads

For the elements in $\mathcal{B}$, which satisfy property (A), we know that equation (37) holds.

Now the relative clause is *non-defining* – it just mentions in passing that all elements in $\mathcal{B}$ satisfy property (A). The main clause states that equation (37) holds for *all* elements in $\mathcal{B}$. See the difference?

2.4 Things you (usually) don't say in English – and what to say instead

Table 2.1 lists some common mistakes and alternatives. The entries should not be taken as gospel – they don't necessarily mean that a given word or formulation is wrong under all circumstances (obviously, this depends a lot on the context). However, in nine out of ten instances the suggested alternative is the better word to use.

Chapter 3

Typography

3.1 Punctuation

Rule 3.1 Use opening (‘) and closing (’) quotation marks correctly.

In $\text{\LaTeX}$, the closing quotation mark is typed like a normal apostrophe, while the opening quotation mark is typed using the French *accent grave* on your keyboard (the *accent grave* is the one going down, as in *frère*).

Note that any punctuation that *semantically* follows quoted speech goes inside the quotes in American English, but outside in Britain. Also, Americans use double quotes first. Oppose

“Using ‘lasers,’ we punch a hole in . . . the Ozone Layer,” Dr. Evil said.

to

‘Using “lasers”, we punch a hole in . . . the Ozone Layer’, Dr. Evil said.

Rule 3.2 Use hyphens (-), en-dashes (–) and em-dashes (—) correctly.

A hyphen is only used in words like ‘well-known’, ‘3-colorable’ etc., or to separate words that continue in the next line (which is known as hyphenation). It is entered as a single ASCII hyphen character (-).

To denote ranges of numbers, chapters, etc., use an en-dash (entered as two ASCII hyphens --) with no spaces on either side. For example, using Equations (1)–(3), we see. . .

As the equivalent of the German *Gedankenstrich*, use an en-dash with spaces on both sides – in the title of Section 2.4, it would be wrong to use a hyphen instead of the dash. (Some English authors use the even longer emdash (—))

instead, which is typed as three subsequent hyphens in $\text{\LaTeX}$. This emdash is used without spaces around it—like so.)

3.2 Spacing

Rule 3.3 Do not add spacing manually.

You should never use the commands `\` (except within tabulars and arrays), `\_` (except to prevent a sentence-ending space after `Dr.` and `such`), `\vspace`, `\hspace`, etc. The choices programmed into $\text{\LaTeX}$ and this style should cover almost all cases. Doing it manually quickly leads to inconsistent spacing, which looks terrible. Note that this list of commands is by no means conclusive.

Rule 3.4 Judiciously insert spacing in maths where it helps.

This directly contradicts Rule 3.3, but in some cases $\text{\TeX}$ fails to correctly decide how much spacing is required. For example, consider

$$f(a,b) = f(a + b, a - b).$$

In such cases, inserting a thin math space `\,` greatly increases readability:

$$f(a,b) = f(a + b, a - b).$$

Along similar lines, there are variations of some symbols with different spacing. For example, Lagrange’s Theorem states that $|G| = [G : H]|H|$, but the proof uses a bijection $f : aH \rightarrow bH$. (Note how the first colon is symmetrically spaced, but the second is not.)

Rule 3.5 Learn when to use `\_` and `\@`.

Unless you use ‘french spacing’, the space at the end of a sentence is slightly larger than the normal interword space.

The rule used by $\text{\TeX}$ is that any space following a period, exclamation mark or question mark is sentence-ending, except for periods preceded by an upper-case letter. Inserting `\` before a space turns it into an interword space, and inserting `\@` before a period makes it sentence-ending. This means you should write

```
Prof.\ Dr.\ A. Steger is a member of CADMO\@.  
If you want to write a thesis with her, you  
should use this template.
```

which turns into

Prof. Dr. A. Steger is a member of CADMO. If you want to write a thesis with her, you should use this template.

The effect becomes more dramatic in lines that are stretched slightly during justification:

Prof. Dr. A. Steger is a member of CADMO. If you

Rule 3.6 Place a non-breaking space (~) right before references.

This is actually a slight simplification of the real rule, which should invoke common sense. Place non-breaking spaces where a line break would look ‘funny’ because it occurs right in the middle of a construction, especially between a reference type (Chapter) and its number.

3.3 Choice of ‘fonts’

Professional typography distinguishes many font attributes, such as family, size, shape, and weight. The choice for sectional divisions and layout elements has been made, but you will still occasionally want to switch to something else to get the reader’s attention. The most important rule is very simple.

Rule 3.7 When emphasising a short bit of text, use `\emph`.

In particular, *never* use bold text (`\textbf`). Italics (or Roman type if used within italics) avoids distracting the eye with the huge blobs of ink in the middle of the text that bold text so quickly introduces.

Occasionally you will need more notation, for example, a consistent typeface used to identify algorithms.

Rule 3.8 Vary one attribute at a time.

For example, for WEIRDSORT we only changed the shape to small caps. Changing two attributes, say, to bold small caps would be excessive ($\text{\LaTeX}$ does not even have this particular variation). The same holds for mathematical notation: the reader can easily distinguish g_n , $G(x)$, $\mathcal{G}$ and G .

Rule 3.9 Never underline or uppercase.

No exceptions to this one, unless you are writing your thesis on a typewriter. Manually. Uphill both ways. In a blizzard.

3.4 Displayed equations

Rule 3.10 Insert paragraph breaks *after* displays only where they belong. Never insert paragraph breaks *before* displays.

L^AT_EX translates sequences of more than one linebreak (i.e., what looks like an empty line in the source code) into a paragraph break in almost all contexts. This also happens before and after displays, where extra spacing is inserted to give a visual indication of the structure. Adding a blank line in these places may look nice in the sources, but compare the resulting display

$$a = b$$

to the following:

$$a = b$$

The first display is surrounded by blank lines, but the second is not. It is bad style to start a paragraph with a display (you should always tell the reader what the display means first), so the rule follows.

Rule 3.11 Never use eqnarray.

It is at the root of most ill-spaced multiline displays. The *amsmath* package provides better alternatives, such as the align family

$$\begin{aligned} f(x) &= \sin x, \\ g(x) &= \cos x, \end{aligned}$$

and multiline which copes with excessively long equations:

$$\begin{aligned} &P[X_{t_0} \in (z_0, z_0 + dz_0], \dots, X_{t_n} \in (z_n, z_n + dz_n)] \\ &= \nu(dz_0) K_{t_1}(z_0, dz_1) K_{t_2-t_1}(z_1, dz_2) \cdots K_{t_n-t_{n-1}}(z_{n-1}, dz_n). \end{aligned}$$

3.5 Floats

By default this style provides floating environments for tables and figures. The general structure should be as follows:

```
\begin{figure}
  \centering
  % content goes here
  \caption{A short caption}
  \label{some-short-label}
\end{figure}
```

Note that the label must follow the caption, otherwise the label will refer to the surrounding section instead. Also note that figures should be captioned at the bottom, and tables at the top.

The whole point of floats is that they, well, *float* to a place where they fit without interrupting the text body. This is a frequent source of confusion and changes; please leave it as is.

Rule 3.12 Do not restrict float movement to only ‘here’ (h).

If you are still tempted, you should avoid the float altogether and just show the figure or table inline, similar to a displayed equation.

Oscillation-Free Discontinuous Galerkin Stabilization

This chapter describes the oscillation-free discontinuous Galerkin (OFDG) stabilization used in this work. The presentation deliberately separates the mathematical operator from its realization in MFEM. The construction combines the projection-based damping mechanism introduced for scalar conservation laws in [4] with the scale normalization, wave-speed scaling, and exact damping evolution developed in [5]. The extension of the original OFDG method to hyperbolic systems is described in [3]. The precise multidimensional sensor implemented here differs from the sensors in those papers; the differences are stated explicitly in Section 4.2.6.

4.1 Mathematical formulation

4.1.1 Purpose of the stabilization

Consider a hyperbolic conservation law, or more specifically the transport problem

$$\partial_t u + \nabla \cdot f(u) = 0, \quad \text{or} \quad \partial_t u + \mathbf{a} \cdot \nabla u = 0. \quad (4.1)$$

Here $u = u(\mathbf{x}, t)$ is the transported scalar quantity, $\mathbf{x} \in \Omega$ is position, t is time, $f(u)$ is its physical flux, and $\mathbf{a}$ is a prescribed transport velocity. The first equation is the conservative form; the second is the special case of linear advection. A system of conservation laws replaces u by a vector of components. The OFDG construction below is applied to each component of that vector.

A discontinuous Galerkin method approximates the solution by a separate polynomial on every mesh element. The polynomials on two neighboring elements are not required to agree at their common face. This is useful because it makes the method local and permits discontinuous physical so-

lutions, but a high-order polynomial can develop nonphysical over- and undershoots close to a shock or another unresolved feature.

OFDG adds damping only to the nonconstant polynomial content of the numerical solution. The amount of damping is determined automatically from jumps between neighboring element polynomials. A smooth, well-resolved solution has small jumps and therefore receives little damping. A discontinuity produces large jumps and therefore stronger damping. The construction has three ingredients:

1. local L^2 projections separate low- and high-order polynomial content;
2. jumps of the solution and its physical derivatives measure a lack of smoothness across element faces;
3. nonnegative damping rates combine the jump measure, element size, solution scale, and normal transport speed.

The following subsections define each ingredient before combining them into the stabilization operator.

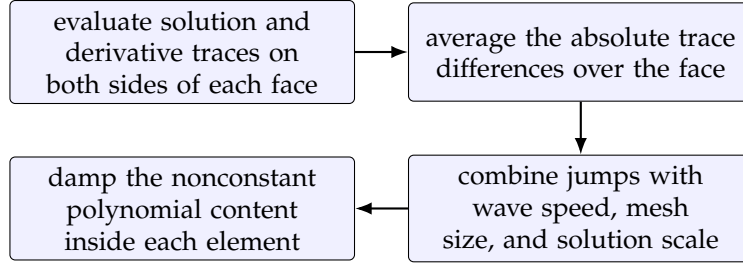


Figure 4.1: Conceptual OFDG pipeline. Information is gathered at element faces, converted into element-local damping rates, and finally applied through local polynomial projections.

4.1.2 Discrete setting

Let $\Omega \subset \mathbb{R}^d$ be partitioned into a mesh $\mathcal{T}_h$, and let $\mathcal{F}_h^i$ denote the set of interior faces. On each element $K \in \mathcal{T}_h$, let $V^k(K)$ be the polynomial space of degree $k \geq 1$ used by the DG discretization. The discontinuous finite element space is

$$V_h^k = \left\{ v \in L^2(\Omega) : v|_K \in V^k(K) \text{ for every } K \in \mathcal{T}_h \right\}. \quad (4.2)$$

The symbol d is the spatial dimension, K denotes one mesh element, and the subscript h indicates that the space is tied to a mesh with a characteristic element size h . An interior face is a full side shared by two elements; an outer boundary side is not included in $\mathcal{F}_h^i$. The notation $v|_K$ means the restriction of v to K . Because that restriction is chosen independently on

every element, a function in V_h^k may have two different values at an interior face.

For a system with m components, the numerical solution belongs to the vector-valued space

$$\mathbf{u}_h = (u_{h,1}, \dots, u_{h,m})^\top \in [V_h^k]^m. \quad (4.3)$$

The present implementation treats these components independently. It does not perform a characteristic decomposition.

On an element K , choose a basis $\{\varphi_i\}_{i=1}^{N_k}$ of $V^k(K)$. A local DG polynomial is represented by

$$u_h|_K(\mathbf{x}) = \sum_{i=1}^{N_k} u_{K,i} \varphi_i(\mathbf{x}). \quad (4.4)$$

The numbers $u_{K,i}$ are the local degrees of freedom. Depending on the basis, they may be nodal values or general expansion coefficients. The theory below is basis independent.

For $0 \leq r \leq k$, denote by $\Pi_K^r : L^2(K) \rightarrow V^r(K)$ the element-local L^2 projection,

$$(\Pi_K^r w - w, v)_K = 0 \quad \text{for every } v \in V^r(K), \quad (4.5)$$

Here $(p, q)_K = \int_K pq \, dx$ is the usual elementwise L^2 inner product. Equation (4.5) says that the projection error is orthogonal to every polynomial of degree at most r . Equivalently, among all functions in $V^r(K)$, $\Pi_K^r w$ is the one closest to w in the root-mean-square sense over K .

We adopt the convention $\Pi_K^{-1} = \Pi_K^0$. The high-pass operators used by the stabilization are

$$\mathcal{R}_K^{(l)} = I - \Pi_K^{l-1}, \quad 0 \leq l \leq k. \quad (4.6)$$

In particular, $\mathcal{R}_K^{(0)} = \mathcal{R}_K^{(1)} = I - \Pi_K^0$. Every $\mathcal{R}_K^{(l)} w$ has zero cell average.

The projection $\Pi_K^r w$ is the best approximation to w in $V^r(K)$ in the $L^2(K)$ norm. It retains polynomial content up to degree r and removes the higher-order content. Consequently, $\mathcal{R}_K^{(l)} w = w - \Pi_K^{l-1} w$ contains precisely the part that the l th damping contribution is allowed to suppress. The convention $\Pi_K^{-1} = \Pi_K^0$ ensures that no damping contribution changes the cell average, which is the constant mode carrying the locally conserved quantity.

4.1.3 Face jumps of physical derivatives

For the multi-index $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, define

$$|\alpha| = \sum_{j=1}^d \alpha_j, \quad \partial^\alpha = \partial_{x_1}^{\alpha_1} \cdots \partial_{x_d}^{\alpha_d}. \quad (4.7)$$

The entries of α count how often differentiation is taken in each coordinate direction, while $|\alpha|$ is the total derivative order. In two dimensions, for example, $\alpha = (1, 0)$ represents ∂_x , $\alpha = (0, 1)$ represents ∂_y , and $\alpha = (1, 1)$ represents ∂_{xy} .

Let $F = \partial K^- \cap \partial K^+$ be an interior face. The sign convention for the jump is immaterial below because only its absolute value occurs; we set

$$[[w]]_F = w^-|_F - w^+|_F. \quad (4.8)$$

The superscripts $-$ and $+$ are labels for the two neighboring elements, not signs. Thus $w^-|_F$ means: approach F from inside K^- and evaluate the polynomial stored on K^- . The trace $w^+|_F$ is obtained independently from K^+ . Even though both traces refer to the same physical points on F , they need not be equal in a DG method.

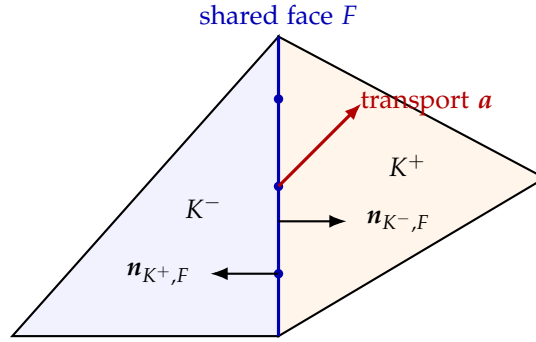


Figure 4.2: Two triangular DG elements sharing the interior face F . The dots indicate representative face quadrature points. At each dot the two element polynomials supply separate traces. Their difference forms the jump. The outward normals point in opposite directions, but the absolute normal wave speed is the same for a single-valued velocity.

For component c and total derivative order l , the implemented face sensor is the sum of normalized mean absolute jumps

$$J_{F,c}^{(l)} = \sum_{|\alpha|=l} \frac{1}{|F|} \int_F |[\partial^\alpha u_{h,c}]_F| \, ds. \quad (4.9)$$

Every symbol in this expression has a direct interpretation:

- F is the shared face shown in Fig. 4.2, and $|F|$ is its length in two dimensions or area in three dimensions;
- $c \in \{1, \dots, m\}$ selects one physical solution component;
- $l \in \{0, \dots, k\}$ is the total derivative order;
- the sum includes every coordinate derivative ∂^α whose total order is l ;

- $\llbracket \partial^\alpha u_{h,c} \rrbracket_F$ is the difference between the two independently evaluated derivative traces; and
- the integral averages the magnitude of that difference over the entire face after division by $|F|$.

Thus $J_{F,c}^{(0)}$ measures the jump of the DG solution itself, while the higher values of l measure all physical partial derivatives of total order l . For example, in two dimensions the $l = 1$ contribution contains the jumps of $\partial_x u_h$ and $\partial_y u_h$, while the $l = 2$ contribution contains those of $\partial_{xx} u_h$, $\partial_{xy} u_h$, and $\partial_{yy} u_h$. The absolute value is applied pointwise before integration, so positive and negative jumps cannot cancel. The sum over all multi-indices ensures that the sensor does not select only one coordinate direction.

Dividing by $|F|$ turns the face integral into a face average. Hence two faces with the same typical jump magnitude produce comparable values even if their physical measures differ. In one spatial dimension a face is a point and (4.9) reduces to the ordinary absolute jump. An earlier version accumulated squared jumps and subsequently took a square root. The current definition uses the intended absolute jump directly; it is both simpler and cheaper to evaluate.

As a simple scalar example, suppose the traces are constant on F with $u_h^-|_F = 2$ and $u_h^+|_F = 1.25$. Then $J_F^{(0)} = |2 - 1.25| = 0.75$, independently of the length of F . If the difference varies along the face, the quadrature described later approximates its mean absolute value instead. This distinction matters: averaging the signed difference could hide a nonsmooth face when positive and negative differences cancel, whereas averaging its absolute value cannot.

Let $\mathbf{a}(\mathbf{x})$ be the prescribed transport velocity and $\mathbf{n}_{K,F}$ the outward unit normal of K on F . The normal wave speed used on the K side of F is

$$\beta_{K,F} = \max_{\mathbf{x} \in F} |\mathbf{a}(\mathbf{x}) \cdot \mathbf{n}_{K,F}|. \quad (4.10)$$

The dot product extracts the part of the velocity crossing the face. Tangential transport has zero normal component and should not determine the exchange between the two neighboring elements. The absolute value makes the rate independent of which side supplies the orientation of the normal. The maximum chooses a safe representative speed when the velocity varies along the face. In the code this maximum is approximated by evaluating the velocity at the face quadrature points. When no velocity is supplied, the implementation uses $\beta_{K,F} = 1$.

4.1.4 Global scale and element damping rates

Following the scale-invariant idea of [5], each component is normalized by one global amplitude. Define the componentwise extrema

$$u_{\min,c} = \min_{K \in \mathcal{T}_h} \min_{x \in K} u_{h,c}(x), \quad u_{\max,c} = \max_{K \in \mathcal{T}_h} \max_{x \in K} u_{h,c}(x). \quad (4.11)$$

The global mean and amplitude are

$$\bar{u}_{h,c} = \frac{1}{|\Omega|} \int_{\Omega} u_{h,c} \, dx, \quad (4.12)$$

$$A_c = \max \{ \bar{u}_{h,c} - u_{\min,c}, u_{\max,c} - \bar{u}_{h,c} \} = \|u_{h,c} - \bar{u}_{h,c}\|_{L^\infty(\Omega)}. \quad (4.13)$$

If $A_c = 0$, all damping rates for component c are defined to be zero. For $A_c > 0$, introduce

$$C_{k,l} = \frac{2l+1}{2(2k-1)l!}, \quad 0 \leq l \leq k. \quad (4.14)$$

The damping rate accumulated for element K , component c , and derivative order l is

$$\delta_{K,c}^{(l)}(u_h) = C_{k,l} \frac{h_K^{l-1}}{A_c} \sum_{\substack{F \subset \partial K \\ F \in \mathcal{F}_h^i}} \beta_{K,F} J_{F,c}^{(l)}. \quad (4.15)$$

For convenient reference, the complete definition is collected below as one system:

$$\begin{aligned} \beta_{K,F} &= \max_{x \in F} |\mathbf{a}(x) \cdot \mathbf{n}_{K,F}|, \\ \delta_{K,c}^{(l)}(u_h) &= \frac{2l+1}{2(2k-1)} \frac{h_K^{l-1}}{l!} \sum_{\substack{F \subset \partial K \\ F \in \mathcal{F}_h^i}} \beta_{K,F} \frac{\sum_{|\alpha|=l} \frac{1}{|\alpha|!} \int_F |[\![\partial^\alpha u_{h,c}]\!]_F| \, ds}{\|u_{h,c} - \bar{u}_{h,c}\|_{L^\infty(\Omega)}}. \end{aligned} \quad (4.16)$$

The symbols can be read as follows:

- K is the element that receives damping, and ∂K is its boundary;
- the face sum runs only over interior faces of K , because only those faces possess two DG traces;
- k is the polynomial degree of the DG space and $l \in \{0, \dots, k\}$ is the derivative order whose jump is being measured;
- c selects one solution component;
- h_K is a characteristic length of element K ; and
- $\mathbf{a} \cdot \mathbf{n}_{K,F}$ is the component of the transport velocity that crosses the face.

The factor h_K^{l-1} consists of the h_K^l derivative scaling of the sensor and the additional h_K^{-1} rate in the damping equation. Consequently, $\delta_{K,c}^{(l)}$ has the dimensions of an inverse time. Equation (4.15) is the exact coefficient computed by the current code; in the implementation it is stored under the name `sigma_elem`.

The terms in (4.15) have distinct roles. The face sum collects information from every interior neighbor of K . The jump $J_{F,c}^{(l)}$ has the physical units of the solution divided by length to the power l . Multiplication by h_K^{l-1} and by the speed $\beta_{K,F}$ converts it to the units of the solution divided by time. Division by A_c leaves an inverse-time damping rate. A larger jump, a faster wave, or a smaller element therefore increases the rate at which unresolved polynomial content is removed.

For a concrete scalar example, take degree $k = 2$, derivative order $l = 0$, one contributing face, $h_K = 0.1$, $A = 1$, normal speed $\beta = 2$, and mean absolute solution jump $J = 0.3$. Since $C_{2,0} = 1/[2(2 \cdot 2 - 1)] = 1/6$, the resulting contribution is

$$\delta_K^{(0)} = \frac{1}{6} \frac{0.1^{-1}}{1} (2)(0.3) = 1. \quad (4.17)$$

If time is measured in seconds, this is a damping rate of one per second. A second interior face would contribute another nonnegative term to the same rate.

The normalization makes the rates invariant under a rescaling and shift of a component. More precisely, away from the zero-amplitude threshold, replacing $u_{h,c}$ by $\lambda u_{h,c} + b$, with $\lambda \neq 0$, leaves every $\delta_{K,c}^{(l)}$ unchanged. Multiplying the transport velocity by a positive factor multiplies the rates by the same factor, so the product of a damping time and a damping rate is unchanged under the corresponding rescaling of physical time.

4.1.5 Projection-based damping operator

The local damping operator is

$$\mathcal{D}_{K,c}(\mathbf{u}_h)u_{h,c} = \sum_{l=0}^k \delta_{K,c}^{(l)}(\mathbf{u}_h) \mathcal{R}_K^{(l)} u_{h,c}. \quad (4.18)$$

Substituting the definition of the high-pass operator gives the same formula in a form that exposes exactly what is removed from the element polynomial:

$$\mathcal{D}_{K,c}(\mathbf{u}_h)u_{h,c} = \sum_{l=0}^k \delta_{K,c}^{(l)}(\mathbf{u}_h) \left(u_{h,c}|_K - \Pi_K^{l-1}(u_{h,c}|_K) \right), \quad \Pi_K^{-1} := \Pi_K^0. \quad (4.19)$$

For each l , the projection reconstructs the part of the element polynomial of degree at most $l - 1$. Subtracting it leaves only higher-degree content. That

remainder is multiplied by the nonnegative rate associated with derivative jumps of order l . Finally, all orders are added. The operation is entirely local once the face-based rates have been computed: no value from a neighboring element is needed while applying (4.19).

When OFDG is used as a semidiscrete stabilization, the ordinary DG operator is augmented by

$$(\partial_t u_{h,c}, v_h)_K = \mathcal{A}_{K,c}(u_h; v_h) - (\mathcal{D}_{K,c}(u_h) u_{h,c}, v_h)_K, \quad (4.20)$$

where $\mathcal{A}_{K,c}$ denotes the unstabilized DG spatial discretization. The function v_h is an arbitrary DG test function, and $(p, q)_K = \int_K p q \, dx$. The minus sign in front of the OFDG term is essential: because the operator is nonnegative, subtracting it removes energy from nonconstant modes instead of amplifying them. The implementation returns the coefficient vector of $\mathcal{D}_{K,c} u_{h,c}$; the caller subtracts this vector from the DG time derivative.

To see which polynomial content is affected, consider one term in (4.18). The $l = 0$ and $l = 1$ terms both act on $u_{h,c} - \Pi_K^0 u_{h,c}$, namely everything except the cell average. For $l \geq 2$, the l th term acts only on content not representable by polynomials of degree $l - 1$. Thus a derivative jump of order l is connected to damping of polynomial modes of degree at least l . The total operator is nonlinear because the rates are recomputed from the current solution, although it is a linear combination of projections once those rates have been frozen.

For example, when $k = 3$, the action of the four rates is:

rate	remainder being multiplied	polynomial content affected
$\delta_{K,c}^{(0)}$	$u_{h,c} - \Pi_K^0 u_{h,c}$	degrees 1, 2, and 3
$\delta_{K,c}^{(1)}$	$u_{h,c} - \Pi_K^0 u_{h,c}$	degrees 1, 2, and 3
$\delta_{K,c}^{(2)}$	$u_{h,c} - \Pi_K^1 u_{h,c}$	degrees 2 and 3
$\delta_{K,c}^{(3)}$	$u_{h,c} - \Pi_K^2 u_{h,c}$	degree 3 only

The constant, degree-zero part never appears in a remainder and is therefore never damped. This is the algebraic reason that the element average is preserved.

The damping term is locally conservative because

$$\int_K \mathcal{R}_K^{(l)} w \, dx = \int_K \left(w - \Pi_K^{l-1} w \right) \, dx = 0 \quad \text{for all } l. \quad (4.21)$$

For $l = 0$, the already stated convention $\Pi_K^{-1} = \Pi_K^0$ is used. The integral vanishes because an L^2 projection onto a space containing constants preserves the element integral of the projected function.

It is also dissipative. Orthogonality of the L^2 projection and nonnegativity of the rates give

$$(\mathcal{D}_{K,c} u_{h,c}, u_{h,c})_K = \sum_{l=0}^k \delta_{K,c}^{(l)} \|u_{h,c} - \Pi_K^{l-1} u_{h,c}\|_{L^2(K)}^2 \geq 0. \quad (4.22)$$

These two identities are direct consequences of the implemented projection operator and do not depend on the particular face sensor.

4.1.6 Exact decay as a modal filter

The same damping operator can be applied in a separate pseudo-time step. Freeze the rates at an input state u_h and solve, independently on each element and for each component,

$$\frac{\partial w_{K,c}}{\partial s} + \sum_{l=0}^k \delta_{K,c}^{(l)}(u_h) \mathcal{R}_K^{(l)} w_{K,c} = 0, \quad w_{K,c}(0) = u_{h,c}|_K. \quad (4.23)$$

Define the mutually orthogonal polynomial shells

$$\mathcal{Q}_K^{(0)} = \Pi_K^0, \quad \mathcal{Q}_K^{(j)} = \Pi_K^j - \Pi_K^{j-1}, \quad 1 \leq j \leq k, \quad (4.24)$$

and their cumulative damping rates

$$\Lambda_{K,c}^{(j)} = \sum_{l=0}^j \delta_{K,c}^{(l)}, \quad 1 \leq j \leq k. \quad (4.25)$$

Since a mode in the j th shell is contained in $\mathcal{R}_K^{(l)}$ precisely when $l \leq j$, the exact solution at pseudo-time τ is

$$w_{K,c}(\tau) = \Pi_K^0 u_{h,c} + \sum_{j=1}^k \exp\left(-\tau \Lambda_{K,c}^{(j)}\right) \mathcal{Q}_K^{(j)} u_{h,c}. \quad (4.26)$$

Eliminating the abbreviations $\Lambda_{K,c}^{(j)}$ and $\mathcal{Q}_K^{(j)}$ gives the directly usable form

$$w_{K,c}(\tau) = \Pi_K^0 u_{h,c} + \sum_{j=1}^k \exp\left(-\tau \sum_{l=0}^j \delta_{K,c}^{(l)}(u_h)\right) \left(\Pi_K^j - \Pi_K^{j-1}\right) u_{h,c}. \quad (4.27)$$

Here j labels a polynomial shell, $\tau \geq 0$ is the duration of the damping step, and the inner sum is the total rate acting on that shell. The exponential factor lies between zero and one: it equals one if the accumulated rate is zero and approaches zero when the rate or damping time becomes large. The first term $\Pi_K^0 u_{h,c}$ is the cell-average mode and has no exponential factor, so it is copied unchanged.

Thus the cell average is unchanged, whereas increasingly high polynomial shells are multiplied by nonnegative decay factors. This is the modal-filter interpretation emphasized in [5]. The exact map avoids a time-step restriction caused solely by stiffness of the frozen damping equation.

The coefficient-space stabilization and the exact decay are two representations of the same operator. Indeed,

$$\sum_{l=0}^k \delta_{K,c}^{(l)} \mathcal{R}_K^{(l)} = \sum_{j=1}^k \Lambda_{K,c}^{(j)} \mathcal{Q}_K^{(j)}. \quad (4.28)$$

The first form is used by `ComputeStabilization`; the exponential of the second form is used by `CompDecay`.

4.2 Implementation decisions and details

4.2.1 Scope and geometric assumptions

The implementation is a standalone MFEM header. It assumes that the finite element space is nonempty and discontinuous, that $k \geq 1$, and that all elements have the same geometry, order, dimension, number of local degrees of freedom, and component layout. For a multicomponent field, the local degrees of freedom of each component must form one contiguous block. All element maps are assumed to be affine, and the mesh must remain unchanged during the lifetime of the OFDG object. These conditions are preconditions rather than properties checked at run time.

Different elements need not have the same Jacobian. They may have different sizes and may be translated, rotated, or affinely skewed. General bilinear quadrilaterals and curved elements are outside the present scope because their inverse Jacobians vary within an element.

More explicitly, MFEM describes a physical element by a map from a standard reference element. The affine assumption means that this map has the form

$$\mathbf{x} = T_K(\boldsymbol{\xi}) = \mathbf{b}_K + J_K \boldsymbol{\xi}, \quad (4.29)$$

where $\boldsymbol{\xi}$ is a reference coordinate, $\mathbf{b}_K$ is a translation, and the matrix J_K is constant within K . The matrix may differ from one element to another. Its scaling, rotation, and shear describe the size, orientation, and skew of the physical element. What the implementation requires is therefore a constant Jacobian on each individual element, not one globally identical Jacobian for the whole mesh. The characteristic length h_K in the damping rate is obtained from MFEM's element-size routine for that element.

4.2.2 Projection matrices

Let $\{\varphi_i\}_{i=1}^{N_k}$ be the local degree- k basis and $\{\psi_a^{(r)}\}_{a=1}^{N_r}$ the degree- r basis. The code assembles

$$(M_k)_{ij} = (\varphi_i, \varphi_j)_K, \quad (M_r)_{ab} = (\psi_a^{(r)}, \psi_b^{(r)})_K, \quad (4.30)$$

$$(B_r)_{ai} = (\psi_a^{(r)}, \varphi_i)_K. \quad (4.31)$$

The matrix representation of $I - \Pi_K^r$ in the degree- k coefficient basis is

$$Q_r = M_k - B_r^\top M_r^{-1} B_r, \quad S_r = M_k^{-1} Q_r = I - M_k^{-1} B_r^\top M_r^{-1} B_r. \quad (4.32)$$

Thus applying S_r to a degree- k coefficient vector directly returns the coefficients of the remainder $w - \Pi_K^r w$. The matrices M_k and M_r are the degree- k and degree- r mass matrices, respectively; B_r couples the two bases; and I is the $N_k \times N_k$ identity matrix. The code stores S_r for $0 \leq r \leq k$.

On an affine element, all three mass and coupling matrices contain the same constant physical Jacobian factor, which cancels in S_r . Therefore the projection matrices are assembled on the first element and reused on every element even when the element Jacobians differ.

4.2.3 Coefficient-to-coefficient derivatives

The repeated derivatives required by (4.9) are represented as polynomial coefficients rather than only as values at quadrature points. At the finite element interpolation points $\{\xi_q\}_{q=1}^{N_k}$, define

$$V_{qi} = \varphi_i(\xi_q), \quad G_{qi}^{(r)} = \frac{\partial \varphi_i}{\partial \xi_r}(\xi_q). \quad (4.33)$$

If $\mathbf{u}$ is a coefficient vector, $G^{(r)} \mathbf{u}$ contains reference-derivative values at the interpolation points. The coefficient vector of that derivative is therefore

$$\mathbf{u}_{\xi_r} = D_{\xi_r} \mathbf{u}, \quad D_{\xi_r} = V^{-1} G^{(r)}. \quad (4.34)$$

This Vandermonde construction is kept explicit in the code. For the affine map $\mathbf{x} = T_K(\xi)$, the physical derivative matrices are

$$D_{x_j}^K = \sum_{r=1}^d (J_K^{-1})_{rj} D_{\xi_r}. \quad (4.35)$$

The inverse Jacobian J_K^{-1} is read once at the reference-element centre and one set of matrices $\{D_{x_j}^K\}_{j=1}^d$ is stored for every element.

All multi-indices with total degree at most k are organized in a directed acyclic graph. Each nonzero multi-index selects one parent of degree one

less and one physical derivative direction. Walking this graph evaluates every required derivative once. The number of stored derivative states per element and component is

$$\sum_{l=0}^k \binom{l+d-1}{d-1} = \binom{k+d}{d}. \quad (4.36)$$

4.2.4 Face quadrature and cached evaluation

MFEM remains the sole source of mesh connectivity. The setup loops over MFEM face numbers and obtains the two adjacent elements from MFEM's face-transformation data. Boundary faces are skipped. For each interior face, a quadrature rule of order $2k+1$ is selected for the MFEM face geometry. The following data are cached directly at the MFEM face index:

$$(E_F^-)_{qi} = \varphi_i(\xi_{F,q}^-), \quad (E_F^+)_{qi} = \varphi_i(\xi_{F,q}^+), \quad \hat{\omega}_q = \frac{\omega_q}{\sum_p \omega_p}. \quad (4.37)$$

For affine faces the physical face Jacobian is constant and cancels between the integral and $|F|$ in (4.9); hence the normalized reference weights $\hat{\omega}_q$ are sufficient. All derivative traces and all components on one side of a face are evaluated together in one dense matrix multiplication. There is no face-pattern matching or deduplication.

If $\mathbf{u}_\alpha^-$ and $\mathbf{u}_\alpha^+$ are the coefficient vectors of the same physical derivative on the two neighboring elements, the actual quadrature calculation is

$$J_{F,c}^{(l)} \approx \sum_{|\alpha|=l} \sum_{q=1}^{N_{q,F}} \hat{\omega}_q \left| (E_F^- \mathbf{u}_\alpha^-)_q - (E_F^+ \mathbf{u}_\alpha^+)_q \right|. \quad (4.38)$$

The implementation therefore performs one subtraction and one absolute value for each derivative, component, and quadrature point. No squared-jump accumulator or final square root is required.

4.2.5 Runtime sequence and storage policy

For every input state, the implementation performs the following operations:

1. evaluate the global mean and scale (4.13) at volume quadrature points;
2. rebuild and store all derivative coefficient states on every element;
3. loop over MFEM interior faces, evaluate the normal wave speeds and the mean absolute jump measures (4.9), and accumulate the rates (4.15) on both neighboring elements;

4. apply either the coefficient-space damping operator (4.18) or the exact decay map (4.26).

The volume mean is integrated with a rule of order $2k + 1$. The extrema used for A_c are the minimum and maximum over the same quadrature values. In floating point arithmetic, the implementation sets the rates of component c to zero when $A_c \leq 10^{-14}$.

There is no bounded or dynamic derivative cache. Every element owns permanent storage for its derivative coefficients, and all of it is rebuilt once for the current state. Likewise, each interior face owns its two evaluation matrices and normalized quadrature weights. Ignoring small container overheads, the principal storage costs are

$$\mathcal{O}(|\mathcal{T}_h| d N_k^2) \quad \text{for physical derivative operators,} \quad (4.39)$$

$$\mathcal{O}\left(|\mathcal{T}_h| m N_k \binom{k+d}{d}\right) \quad \text{for derivative coefficient states,} \quad (4.40)$$

$$\mathcal{O}\left(\sum_{F \in \mathcal{F}_h^i} N_{q,F} (2N_k + 1)\right) \quad \text{for face evaluation data.} \quad (4.41)$$

This policy uses more predictable memory than an evicting cache and removes cache lookups and replacement decisions from the face loop.

An optional active-element mask can restrict stabilization or decay to selected elements. A face is omitted only when neither neighbor is active, so an active element still receives contributions from every adjacent interior face. Runtime scratch arrays and optional internal timers are owned by the OFDG object. Because that scratch state is mutable and reused, concurrent calls on the same object are not supported.

4.2.6 Relation to the published formulations

The original multidimensional OFDG sensors in [4, 3] measure derivative jumps at element vertices. The system formulation in [3] additionally transforms jumps to characteristic variables and takes a maximum over characteristic components. The current implementation instead uses the face-averaged absolute-jump measure (4.9) and constructs a separate rate for each physical component. Its global amplitude normalization, normal wave-speed factor, and exact decay map are motivated by the later OEDG formulation [5].

Consequently, the implementation should be regarded as the specific hybrid OFDG and OEDG damping variant defined by Eq. (4.9), Eq. (4.15), and Eq. (4.18), not as a literal reproduction of one published algorithm. Conservation and dissipation follow directly from Eq. (4.21) and Eq. (4.22);

convergence results proved for the published vertex-based or characteristic-variable sensors do not, by themselves, constitute a proof for this modified sensor.

The KXRCF troubled-cell indicator is intentionally not included in this chapter. Its eventual role is to provide the active-element set used by the existing mask interface; its definition and interaction with OFDG will be documented after the KXRCF implementation has been finalized.

Appendix A

Dummy Appendix

You can defer lengthy calculations that would otherwise only interrupt the flow of your thesis to an appendix.

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Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

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